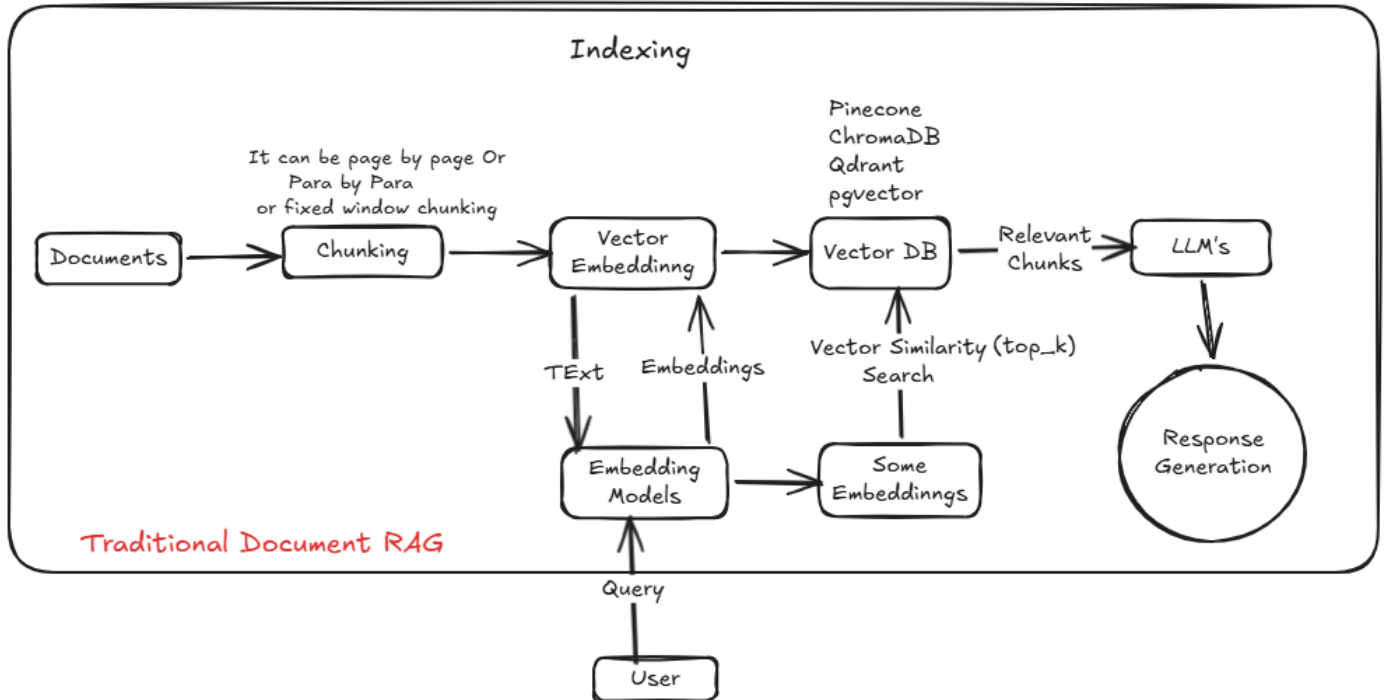


Research About Knowledge Base –

Types of RAG we can implement –

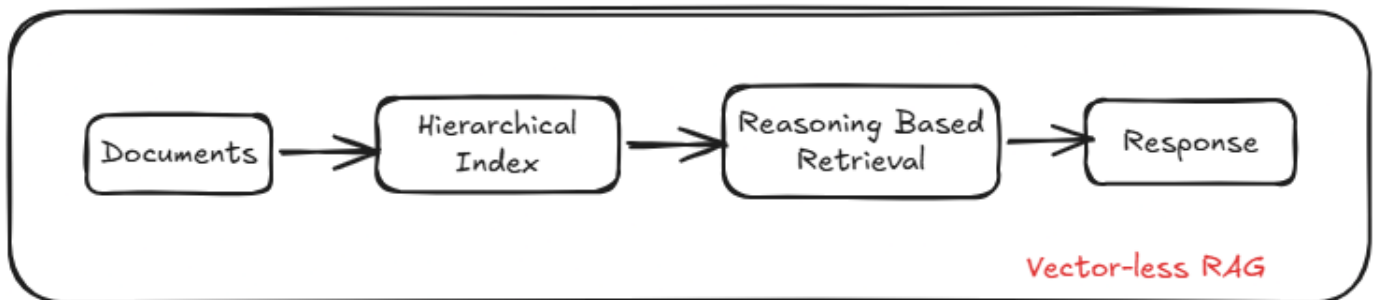
1 . Normal RAG



Drawbacks –

- 1 . Blindly Chunking - We just randomly do chunking without any context that may lead to loss of information (Full context of any para or any context may loss)
- 2 . Relational Data Context is Lost
- 3 . If prompt is not accurate (Not contain proper keywords) the model may fail

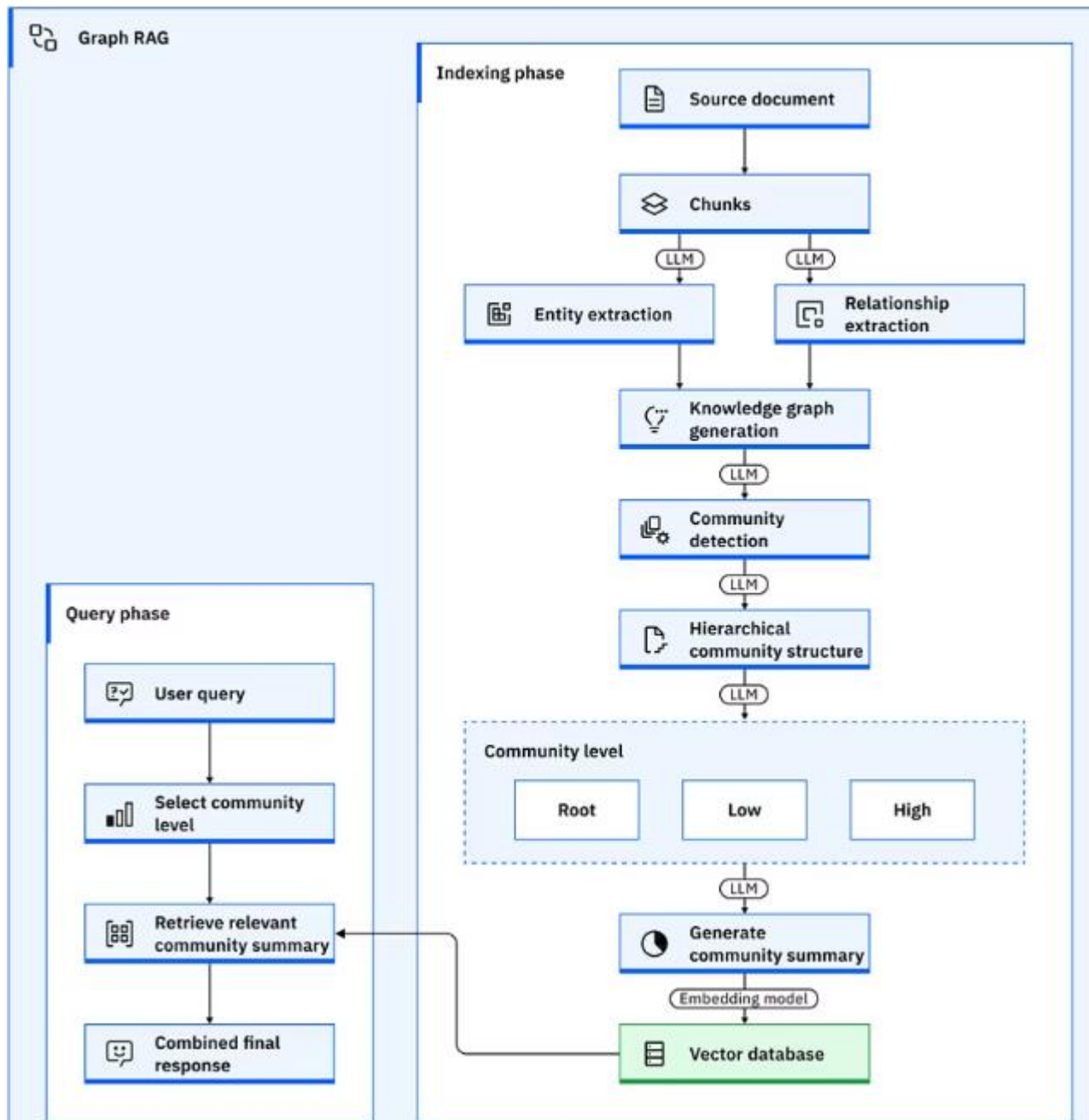
2 . Vectorless RAG (PageIndex)



What's Better –

- 1 . Instead of semantic similarity search it builds a hierarchical table of contents (TOC) tree from a document and uses LLM (reasoning model) to reason over that structure
- 2 . Model first identifies the most relevant section using document hierarchy then navigates to that section to generate a precise , cited answer
- 3 . It only picks up the relevant nodes and traverse only over the relevant child nodes , which further helps in providing response with proper context

3 . Graph RAG

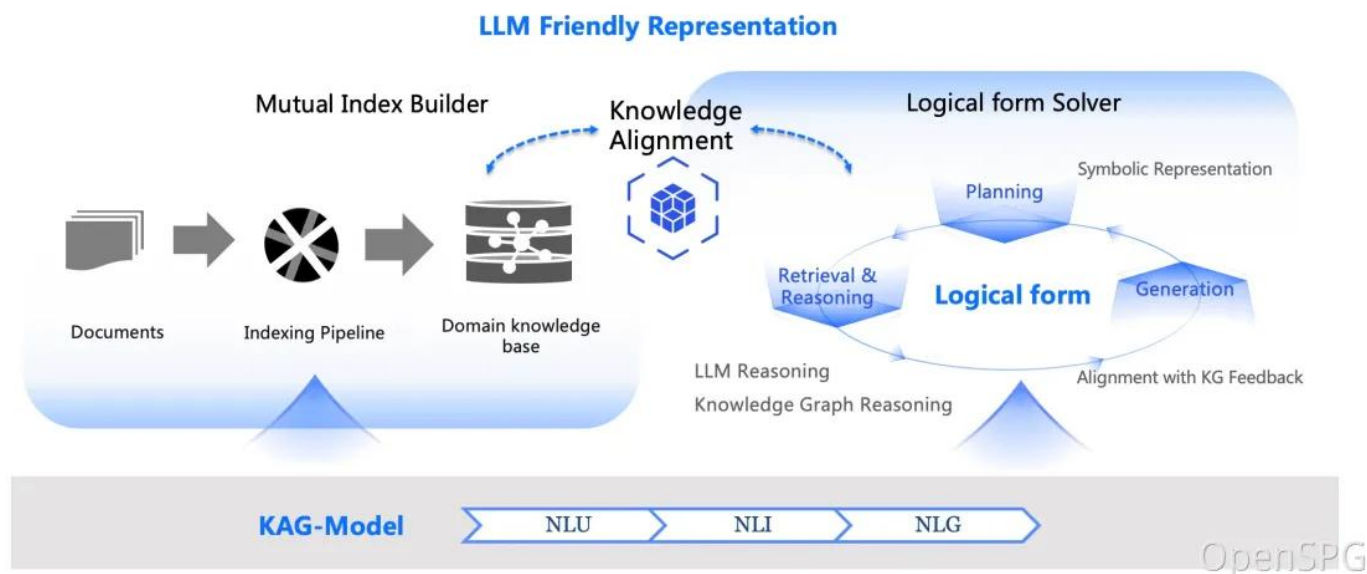


What's better -

1. Instead of semantic similarity or hierarchical TOC, it builds a knowledge graph by automatically extracting entities (people, concepts, organizations, events, etc.) and **explicit relationships** between them. It then uses this graph structure + LLM reasoning to answer questions that require understanding connections, not just keyword or semantic matches.
2. Model first identifies relevant entities and their neighboring relationships in the graph, then intelligently traverses the graph (multi-hop) to gather connected information. This allows it to follow logical chains ($A \rightarrow B \rightarrow C \rightarrow D$) and generate more comprehensive, accurate, and contextually rich answers.
3. It only retrieves and reasons over the relevant subgraph (connected nodes and edges) instead of pulling isolated text chunks. This focused traversal provides precise, highly relevant context while maintaining clear relationships, leading to better multi-hop reasoning and significantly fewer hallucinations on complex queries.
4. Graph RAG excels at global understanding and explainability — it creates hierarchical community summaries of the entire dataset (or document collection) and can trace the exact reasoning path

through visible entity-relationship chains. This makes answers more transparent, auditable, and trustworthy compared to black-box vector similarity.

4 . Knowledge Augmented Generation (KAG)



What's Better -

1. Instead of pure vector similarity (traditional RAG) or basic graph traversal (Graph RAG), KAG builds a high-quality, domain-specific knowledge graph combined with vector retrieval. It uses LLM-friendly knowledge representation and mutual-indexing between the graph and original text chunks for more precise and relevant knowledge access.
2. It introduces a logical-form-guided hybrid reasoning engine that understands and respects domain-specific logic (numerical values, temporal relations, expert rules, hierarchies, etc.). The system first converts the query into a logical form, then performs hybrid reasoning over both the knowledge graph and vector space.
3. KAG enables bidirectional enhancement between the LLM and the Knowledge Graph. The LLM helps improve knowledge alignment and graph quality through semantic reasoning, while the structured graph provides the LLM with precise, logical context — resulting in significantly more professional, accurate, and logically consistent answers.
4. It excels at professional and multi-hop reasoning tasks by reducing the gap between semantic similarity and true knowledge relevance. This leads to much lower hallucination rates and higher factual accuracy in complex domain-specific scenarios compared to standard RAG or Graph RAG.

What we can implement – Hybrid approach

- a. **Lightweight and Memory-Efficient Design** - Instead of building heavy knowledge graphs or preloading entire documents into cache, this architecture uses optimized vector search with strong metadata filtering. It keeps memory usage low (fits comfortably within 16GB RAM limit) while allowing the robot to run navigation and other processes smoothly.
- b. **Smart & Flexible Ingestion Pipeline** - Documents are automatically parsed with structure-aware techniques (headings, sections, pages). It supports configurable chunking strategies (recursive, fixed-size, or heading-based) and automatically captures rich metadata (filename, upload date, author, page numbers, section titles, custom tags). This results in better context preservation compared to blind chunking in normal RAG.

- c. Efficient and Accurate Retrieval - The system first applies metadata filters (document, date, tags, sections) and then performs vector similarity search only on the relevant subset. This hybrid retrieval approach reduces noise, retrieves more precise chunks, and provides rich context (including source document, page, and section) for the LLM to generate accurate, well-cited answers.
- d. It supports easy document management (upload, preview, re-index, delete, search), model selection (different embedding/LLM models), multilingual documents (Hindi+English), and incremental indexing. The architecture works efficiently with Ollama on low-resource hardware while delivering fast responses with proper document context for the robot to speak naturally.

